A Steady-State Groundater Model for the Heathcote Valley

Objectives

Set up a steady-state groundwater model of the Heathcote Valley. We will extend this model in next week's tutorial and you will also need to use it for the Assignment.

Resources

These instructions and this week's lectures on using Model Muse to set up a MODFLOW model. A png image extracted from Google Maps with a few highlighted features.

Instructions

Open and view the file named heathcote_topo_4x4km_annotated.png. This screen grab from Google Maps outlines the 4×4 km region we will develop a model for. Note the locations of two key elements:

- Topographic contours. These are highlighted by thin brown lines. The 320, 200, 40 and 0 m (sea level) contours are shown.
- Gravel channel. An old river bed comprising a well-defined region of open framework gravels that is embedded in the lowermost confined aquifer. Thick purple hatching defines this feature.

There are also some wells indicated. We'll use these in a later session.

<u>Model grid</u>

We will start by developing the model grid. Open Model Muse and create a new model. We'll conceptualise the Heathcote valley as a simple system with three layers:

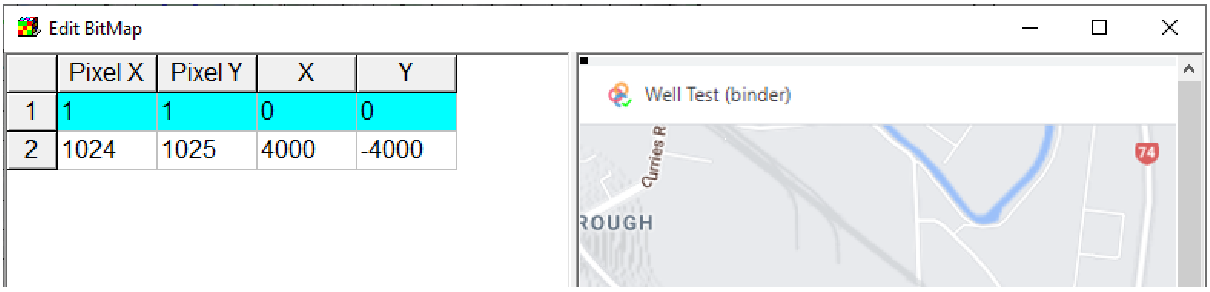
1. Upper unconfined aquifer, extending from the ground surface to 5 m below sea level.
2. A clay confining unit, approximately 55 m thick.
3. A confined gravel/sand aquifer, approximately 30 m thick, through which the gravel channel passes.

Create a 4×4 km model with three layers and 100×100 m area grid blocks. Set layer elevations relative to 0 m at sea level.

We want to reference our model to the annotated image of the Heathcote Valley. We can load the image and then set it as the model backdrop as we are defining objects, constraints and boundary conditions.

File > Import > Image > heathcote_topo_4x4km_annotated.png

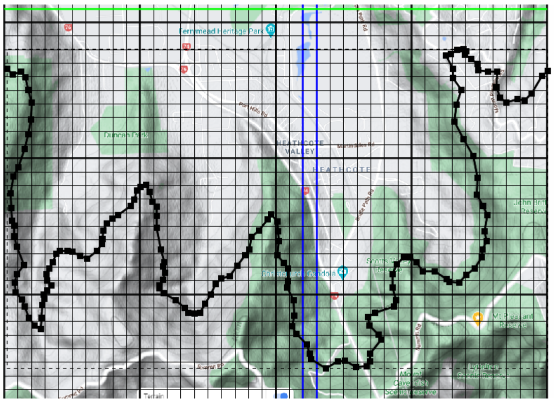
Add a second row to define the mapping. Fill in the rows so that (image) pixel locations match (model) X and Y coordinates.

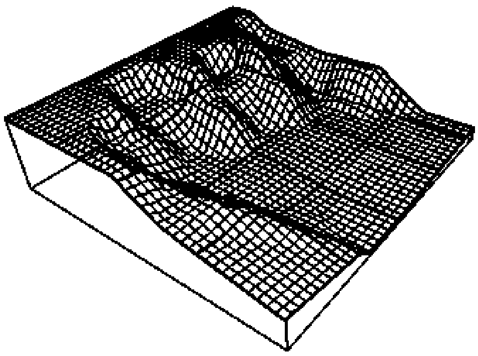


	Pixel X	Pixel Y	X	Y
1	1	1	0	0
2	1024	1025	4000	-4000

Next, we'll assign topography to the model. This will be a little more complex than the constant slope considered in lectures but the same principles apply:

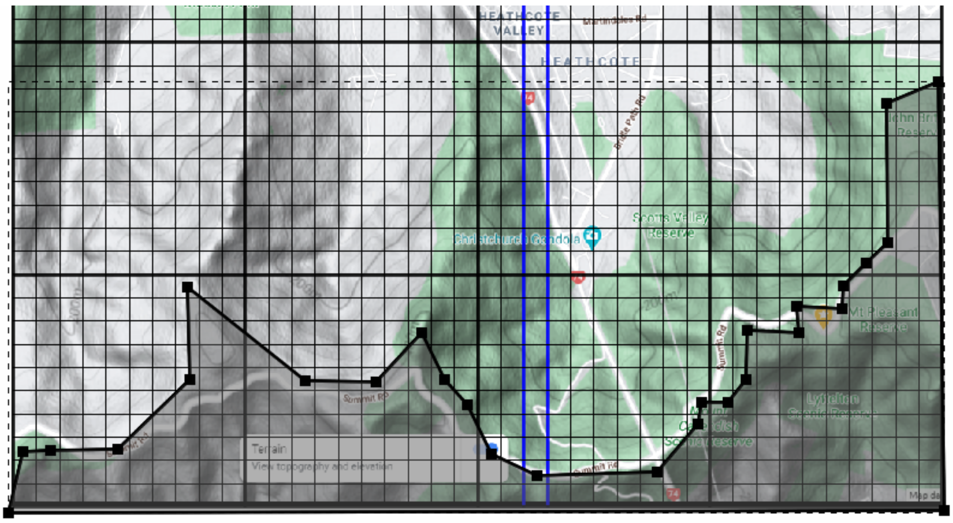
1. ***Turn on Model_Top interpolation in Data > Edit Datasets… > Required > Layer Definition > Model_Top > Interpolation > Triangle Interp.***
2. ***Create a polyline that traces the 200 m contour and then, setting values of cells by interpolation, assign this object's Required > Layer Definition > Model_Top > Formula = 200***
3. ***Create similar polylines for the 320, 40 and 0 m contours and verify that a plausible looking topography has been generated.***





We don't want any model cells south of the 320 m contour contributing to the model. This region corresponds to a different catchment and we're assuming no groundwater flow between them.

Trace a polygon that follows the 320 m contour and then drop down and across to enclose all cells to the south. Set Z formulas to 2 and set higher and lower Z-coordinates to enclose the entire model (Model_Top and Aquifer_Bottom, respectively). Then, Data Sets > Required > Hydrology > Active > Formula = False



Hydrologic properties

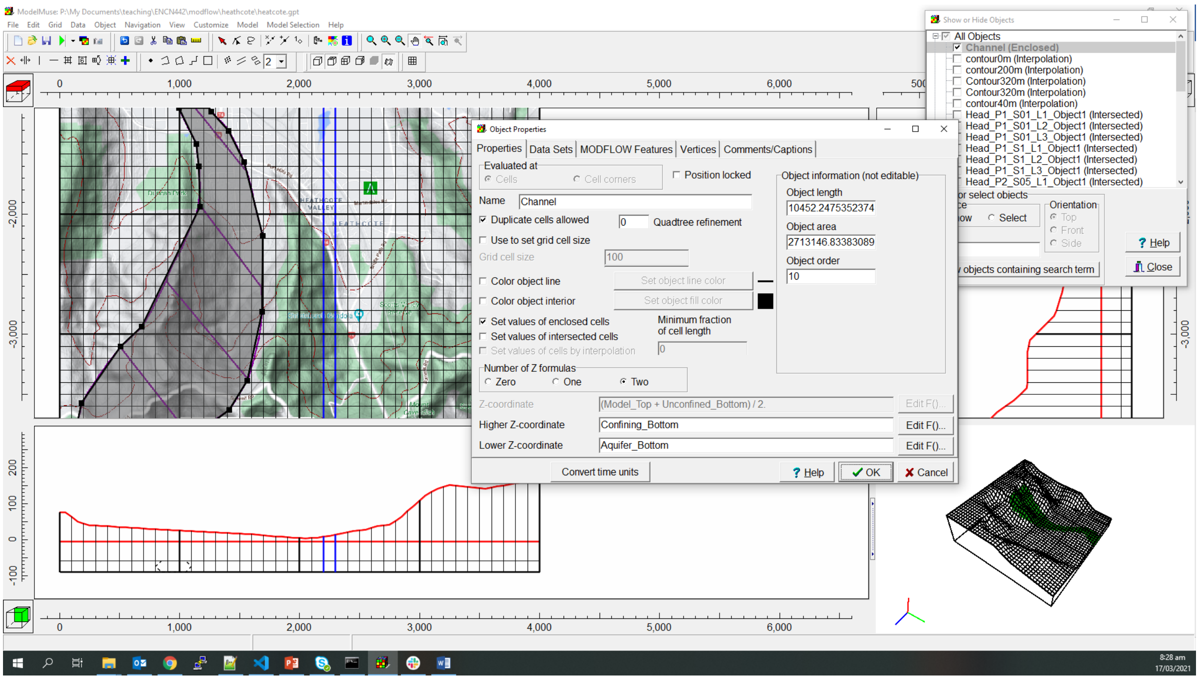
The three layers – unconfined, confining and aquifer – have the following hydraulic conductivities: K_x = 0.0002, 0.0001 and 0.001 m/s, respectively. $K_y = K_x$ and $K_z = K_x/10$ as usual.

Assign global hydraulic conductivities to the three layers via Data > Edit Data Sets…

Change the Cell_Type in the top layer so that calculations there use the saturated thickness.

The gravel channel mapped in the annotated image is a region of extra high conductivity within the aquifer. We can define a region corresponding to this channel and then assign it a higher value of K_x that overwrites the background value set above.

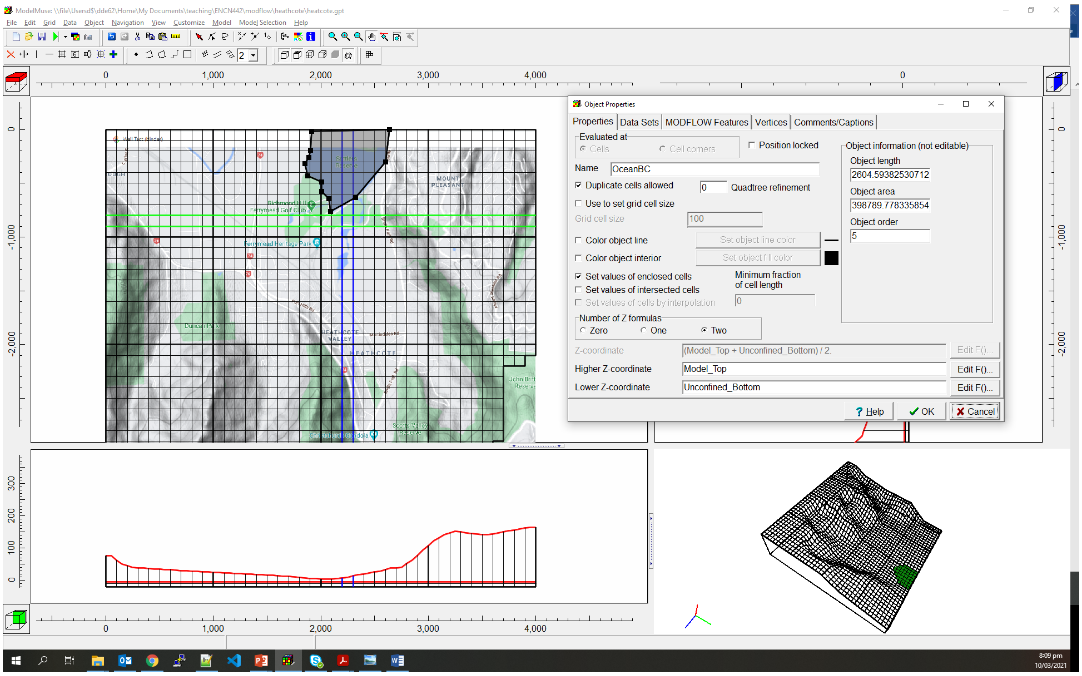
Create a polygon that encloses the channel. Set its vertical constraints to select the Aquifer layer only. In Data Sets, assign it K_x = 1 m/s, with K_y and K_z defined as usual.



Boundary conditions

There are two important boundary conditions in this model: (1) a constant head boundary condition representing the ocean (Settlers Reserve, enclosed by the 0 m contour), and (2) rainfall recharge over the top surface equal to 20 mm/month.

1. ***In Model > MODFLOW Time, ensure there is one Steady state stress period of length 1 s (the length doesn't actually matter but 1 s will be helpful later.)***
2. ***Turn on RCH and CHD packages by selecting the appropriate checkboxes in Model > MODFLOW Packages and Programs > Boundary conditions.***
3. ***Draw a polygon that encloses the blue body of water. Add constraints to select enclosed cells in the uppermost (unconfined) layer. Set its MODFLOW Features > CHD so that starting and ending head are both 0.***
4. ***Draw a polygon that encloses the entire model. Add constraints to select enclosed cells in the uppermost layer. Set its MODFLOW Features > RCH to reflect rainfall of 20 mm/month***



Running the model

Run the model and debug any problems (get in touch with me if you get stuck).

Visualise the steady-state head distribution in the model by File > Import > Model Results… (set the Contour Interval to 0.1 to get a good view of the head distribution).

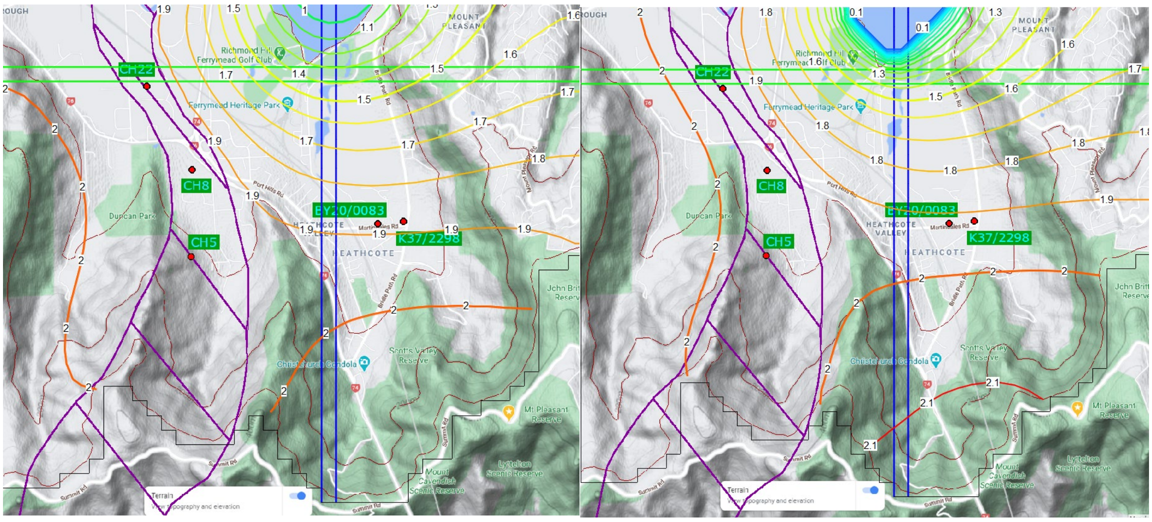


Figure: Contours of head in the uppermost (left) and lowermost layers (right)

Can you see the effect of the gravel channel in the head distribution? Change its conductivity so it is no longer distinct from the aquifer background, rerun the model and then compare contours. How has it changed?

Is the head in the aquifer *beneath* the ocean zero, the same as the unconfined layer? Why or why not?